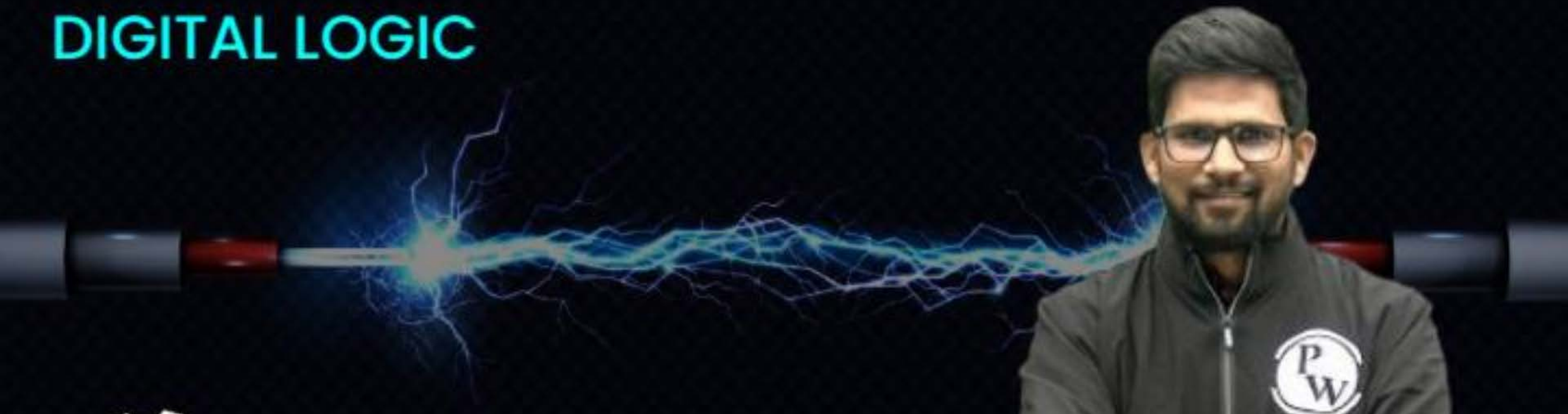
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 07

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture



- Asynch Counter
-
-



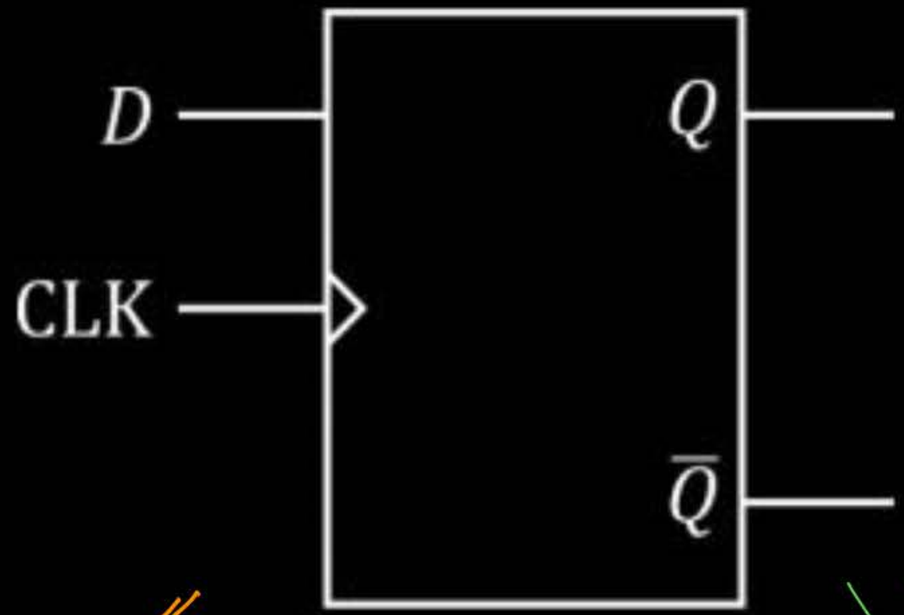
- D-FF, T-FF
- Synch. Counter design.

D-FF and T-FF

$Q(n+1) = D$

- What is D-FF ?

Transparent latch



D	Q(n)	Q(n+1)
0	0	0
0	1	0
1	0	1
1	1	1

$Q(n+1)[D, Q(n)] = \Sigma 2, 3$

$Q(n+1) = D$.

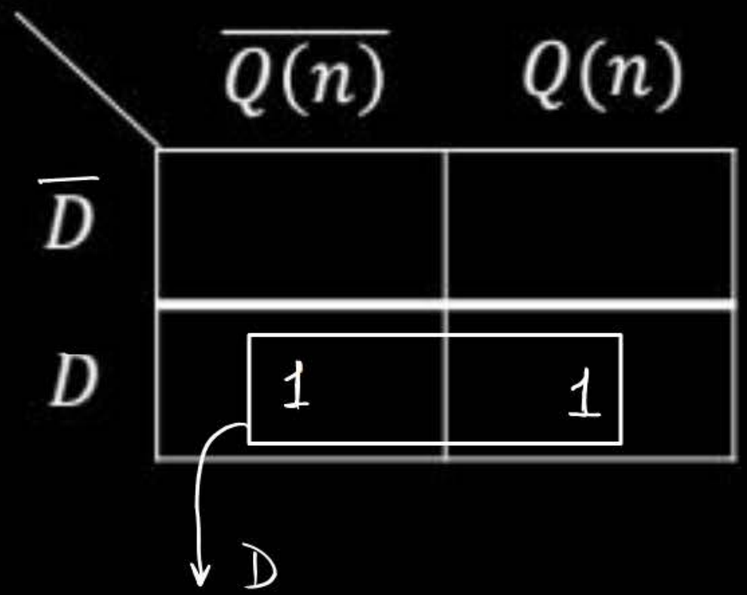
	$\overline{Q(n)}$	$Q(n)$
$\overline{D}$		
D	1	1

$\rightarrow D$.

- Characteristic table :

D	$Q(n)$	$Q(n+1)$
0	0	0
0	1	0
1	0	1
1	1	1

$\Rightarrow$



$Q(n+1) = D$

- Excitation table :

$Q(n)$	$Q(n+1)$	D
0	0	0
0	1	1
1	0	0
1	1	1

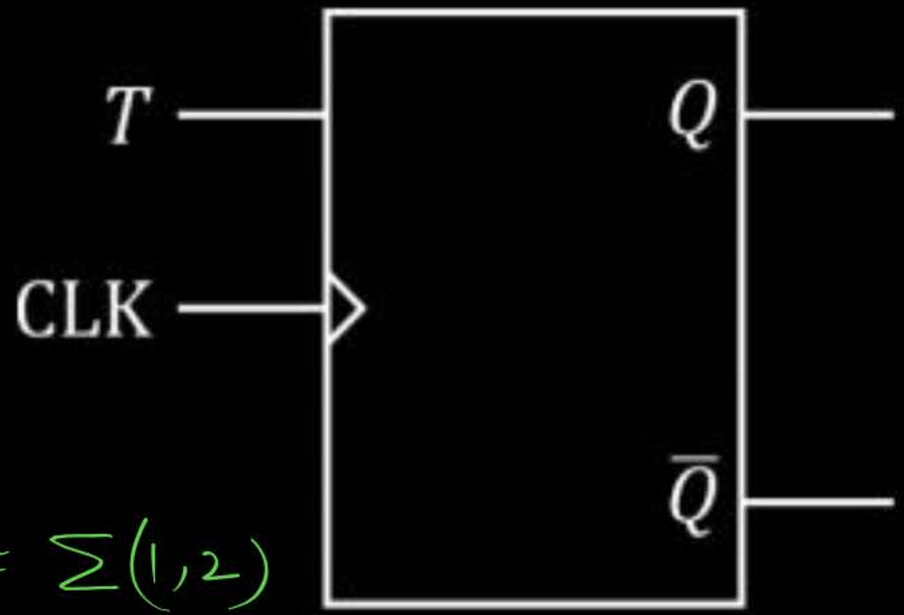
T-FF

- What is T-FF ?

$Q(n+1) = T \oplus Q(n)$

$T=1 \longrightarrow Q(n+1) = 1 \oplus Q(n) = \overline{Q}(n)$

$T=0 \longrightarrow Q(n+1) = 0 \oplus Q(n) = Q(n)$



$Q(n+1)\,[T, Q(n)] = \Sigma(1,2)$

$= \overline{T}\,Q(n) + T\,\overline{Q}(n)$

$\boxed{Q(n+1) = T \oplus Q(n)}$

T	Q(n)	Q(n+1)
0	0	0
0	1	1
1	0	1
1	1	0

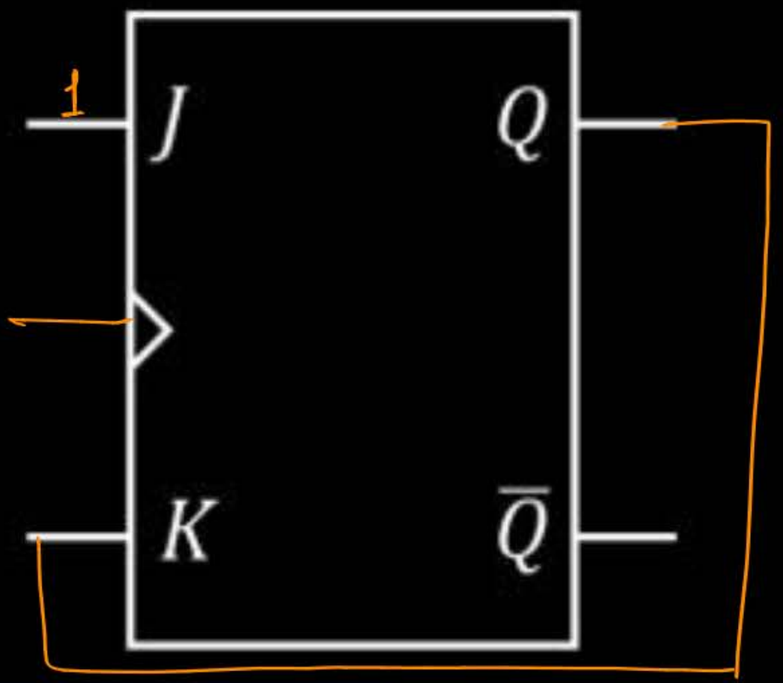
Characteristic table.

- Excitation table :

$Q(n)$	$Q(n+1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

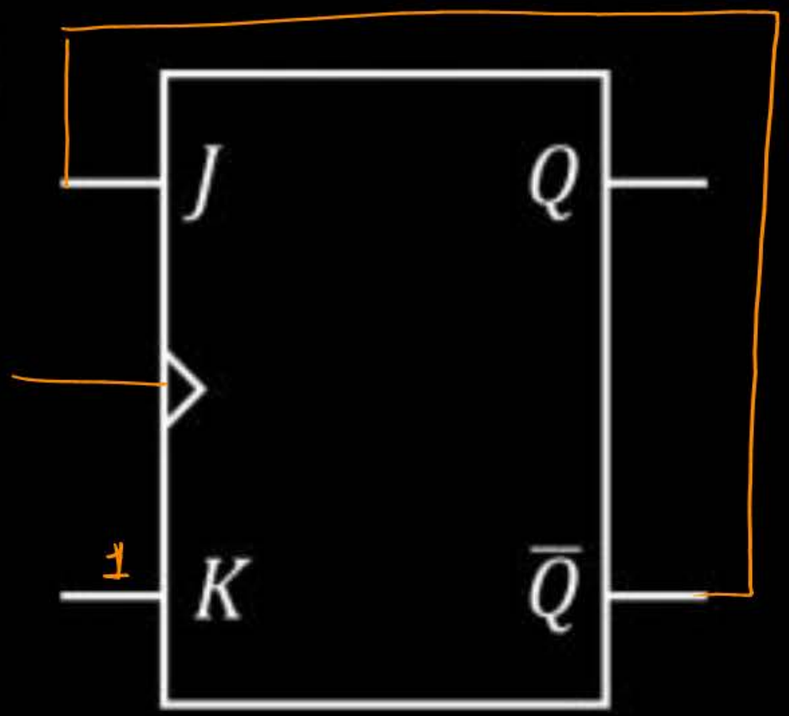
Toggle mode of operation

(i)



$Q(n+1) = \overline{Q}(n)$

(ii)



$Q(n+1) = \overline{Q}(n) \cdot \overline{Q}(n)$

$Q(n+1) = \begin{matrix} +\overline{1}\,Q(n) \\ \overline{Q}(n) \end{matrix}$

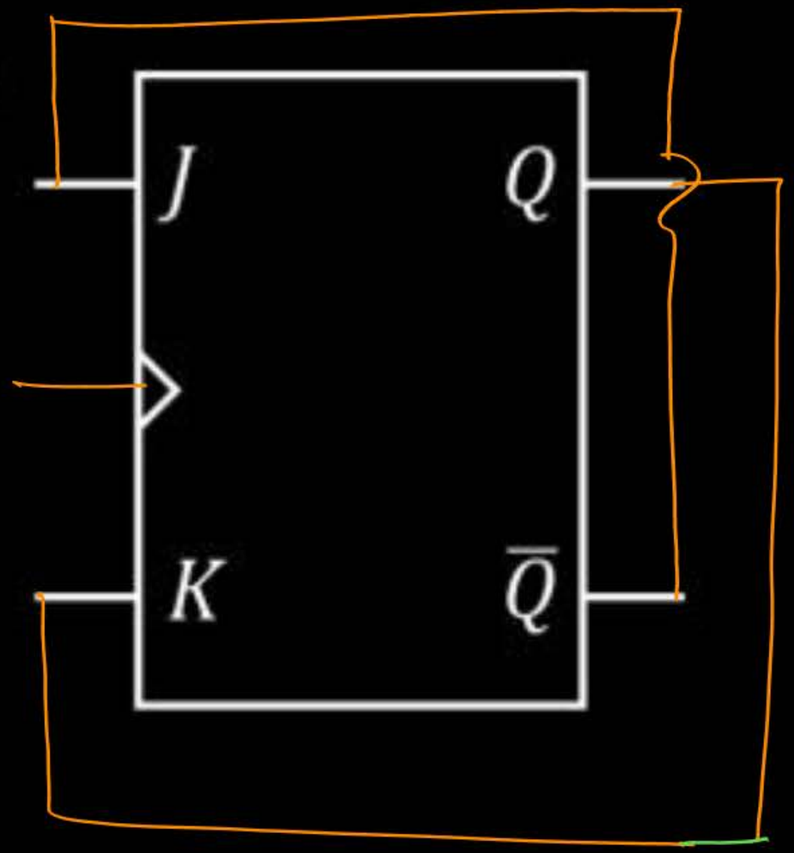
$Q(n+1) = \overline{J}\,\overline{Q}(n) + \overline{K}\,Q(n) \longrightarrow J=1, K=1 \quad Q(n+1) = \overline{Q}(n)$

$[1, \overline{Q}(n)] \qquad [1, Q(n)]$

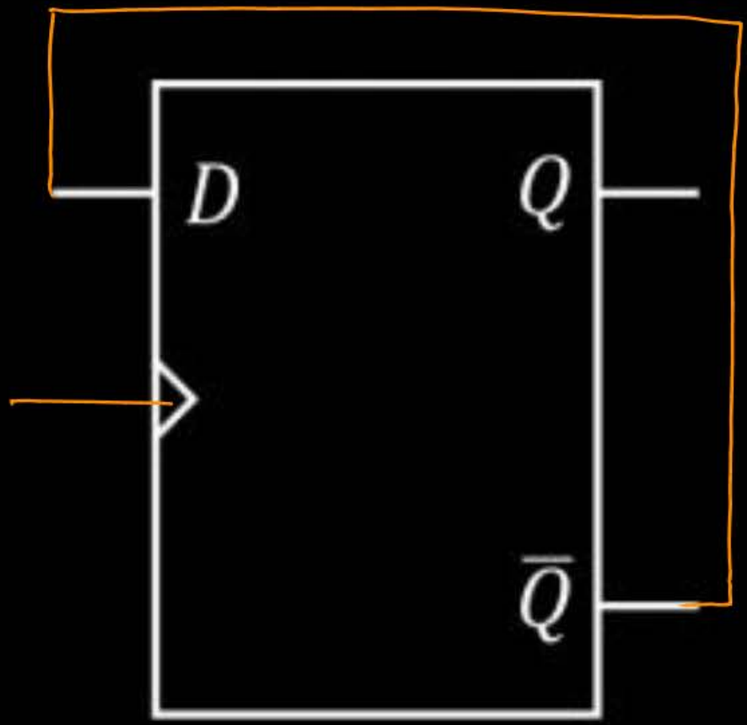
$Q(n+1) = D$

$Q(n+1) = \overline{Q}(n)$

(iii)

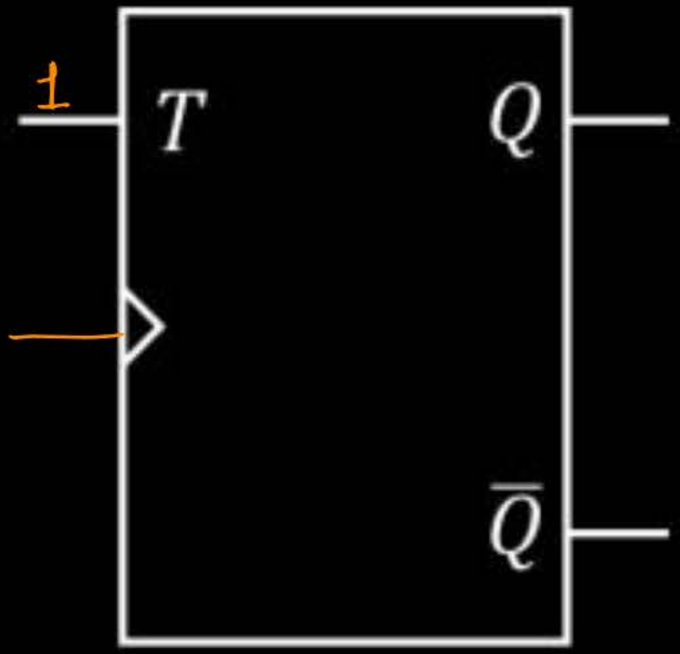


(iv)



$$Q(n+1) = J\overline{Q}(n) + \overline{K}Q(n)$$
$$= \overline{Q}(n) \cdot \overline{Q}(n) + \overline{Q}(n) \cdot Q(n)$$
$$= \overline{Q}(n)$$

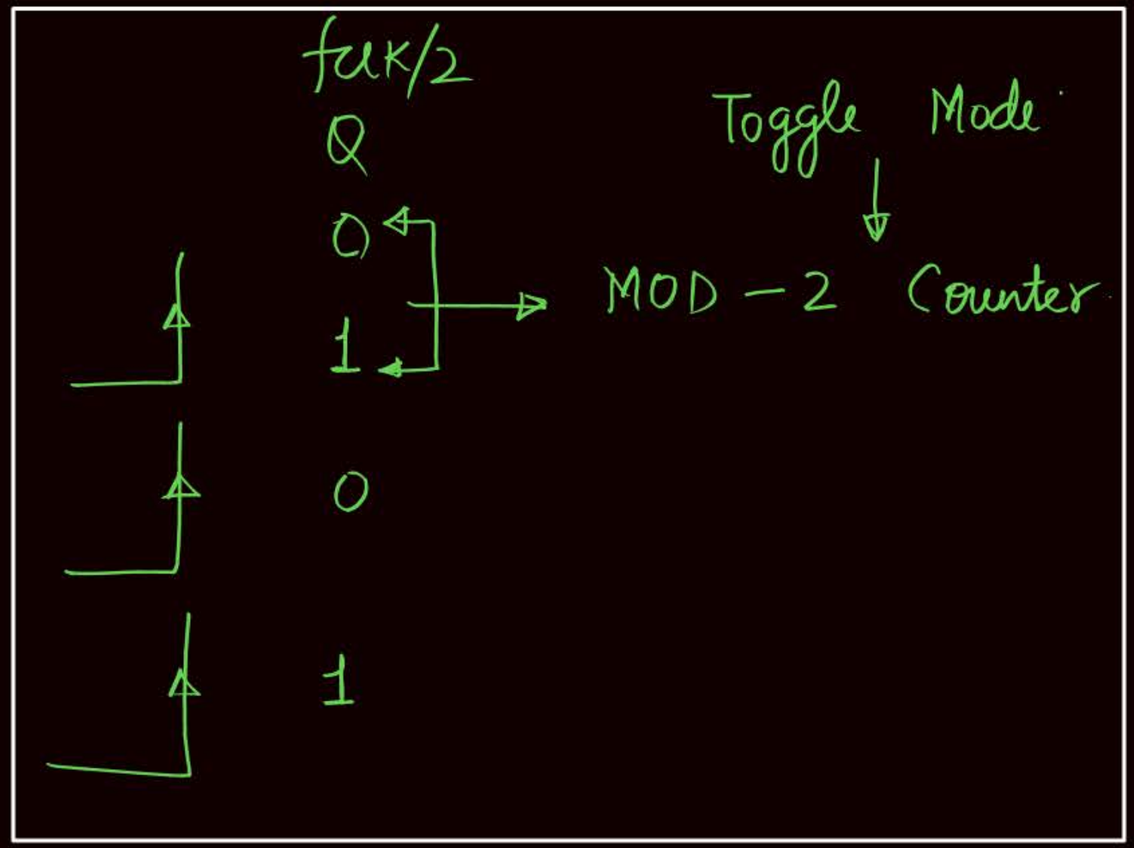
(v)



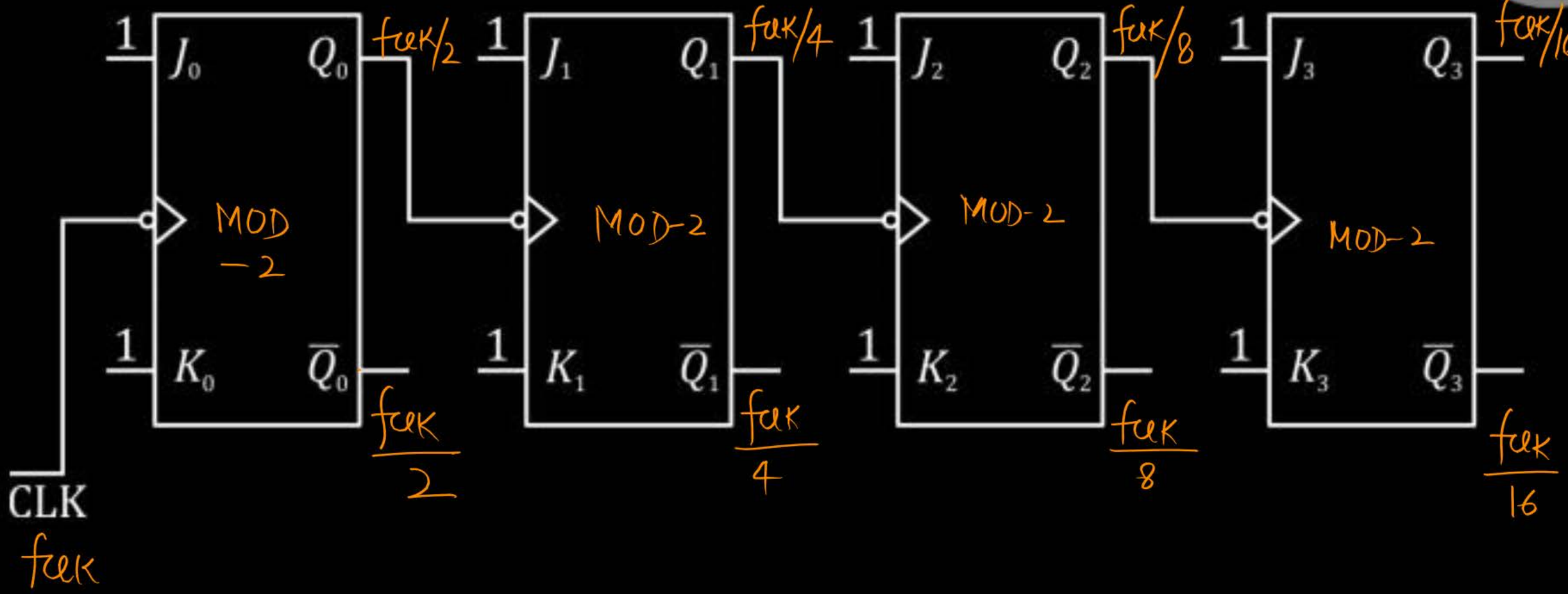
$Q(n+1) = T \oplus Q(n) = 1 \oplus Q(n)$

$Q(n+1) = \overline{Q}(n)$.

$$Q(n+1) = \overline{Q}(n)$$



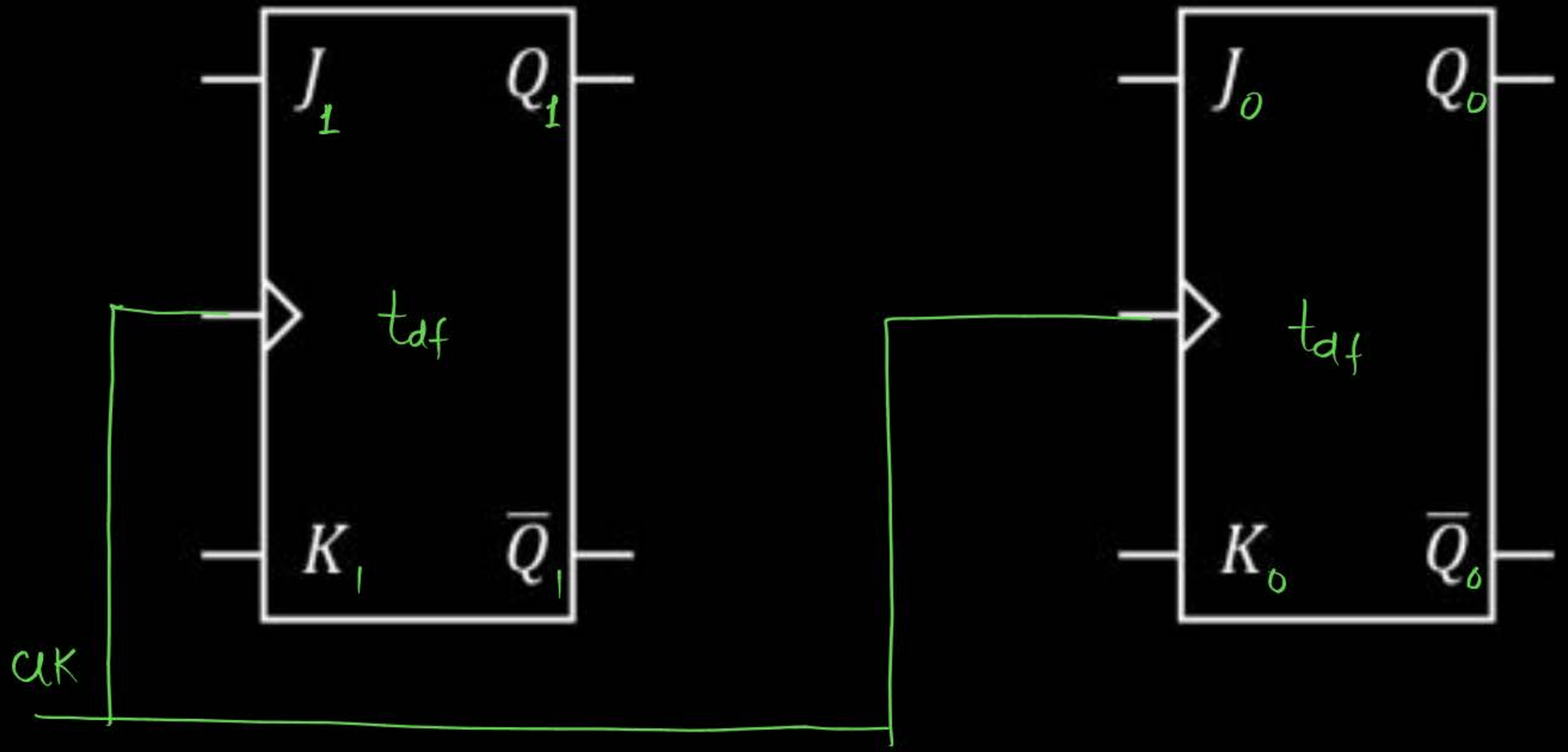
Lets analyze asynchronous counter :

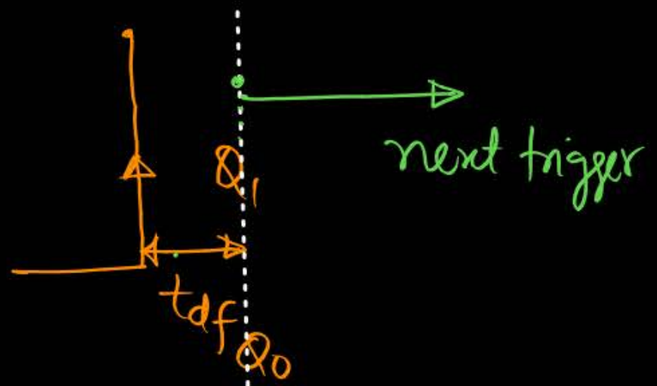


Synchronous Counter

- What is synchronous counter ?
  - Every ff used is driven by same external clock.

- Lets take a synchronous counter and analyze it :





for proper operation.

$T_{clk} \geq t_{df} \rightarrow (T_{clk})_{min} = t_{df}$

$f_{clk} \leq \frac{1}{t_{df}}$

$(f_{clk})_{max} = 1/t_{df}$

How to design a Synchronous Counter

- 0 – 2 – 3 – 1 — MOD-4 Counter ⟶ $n = 2$,

  Design above counter using JK-FF

$Q_1\ Q_0$

$0\ \ 0$ ⟶ $(10)_2$ ⟶ $(11)_2$ ⟶ $(01)_2$ ⟶ $(00)_2$

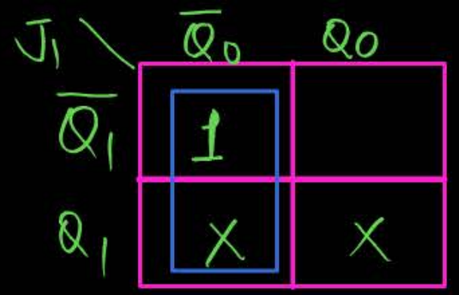
Q_1	Q_0	Q_1^+	Q_0^+	J_1	K_1	J_0	K_0
0	0	1	0	1	X	0	X
0	1	0	0	0	X	X	1
1	0	1	1	X	0	1	X
1	1	0	1	X	1	X	0

$$J_1[Q_1(n), Q_0(n)] = \Sigma 0 + d\Sigma(2,3)$$

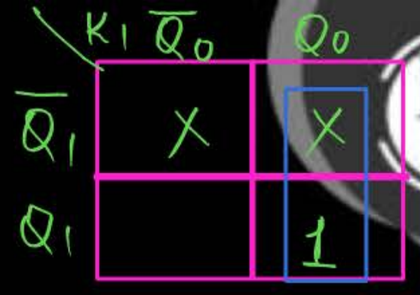
$$K_1[Q_1(n), Q_0(n)] = \Sigma 3 + d\Sigma(0,1)$$

$$J_0[Q_1(n), Q_0(n)] = \Sigma 2 + \Sigma(1,3)$$

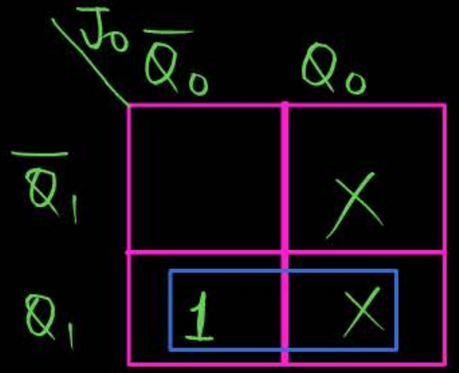
$$K_0[Q_1(n), Q_0(n)] = \Sigma 1 + \Sigma(0,2)$$



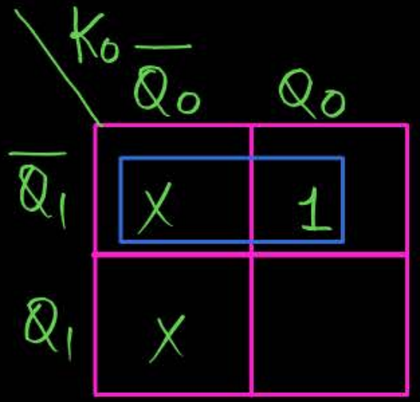
$J_1 = \overline{Q_0}(n)$



$K_1 = Q_0(n)$

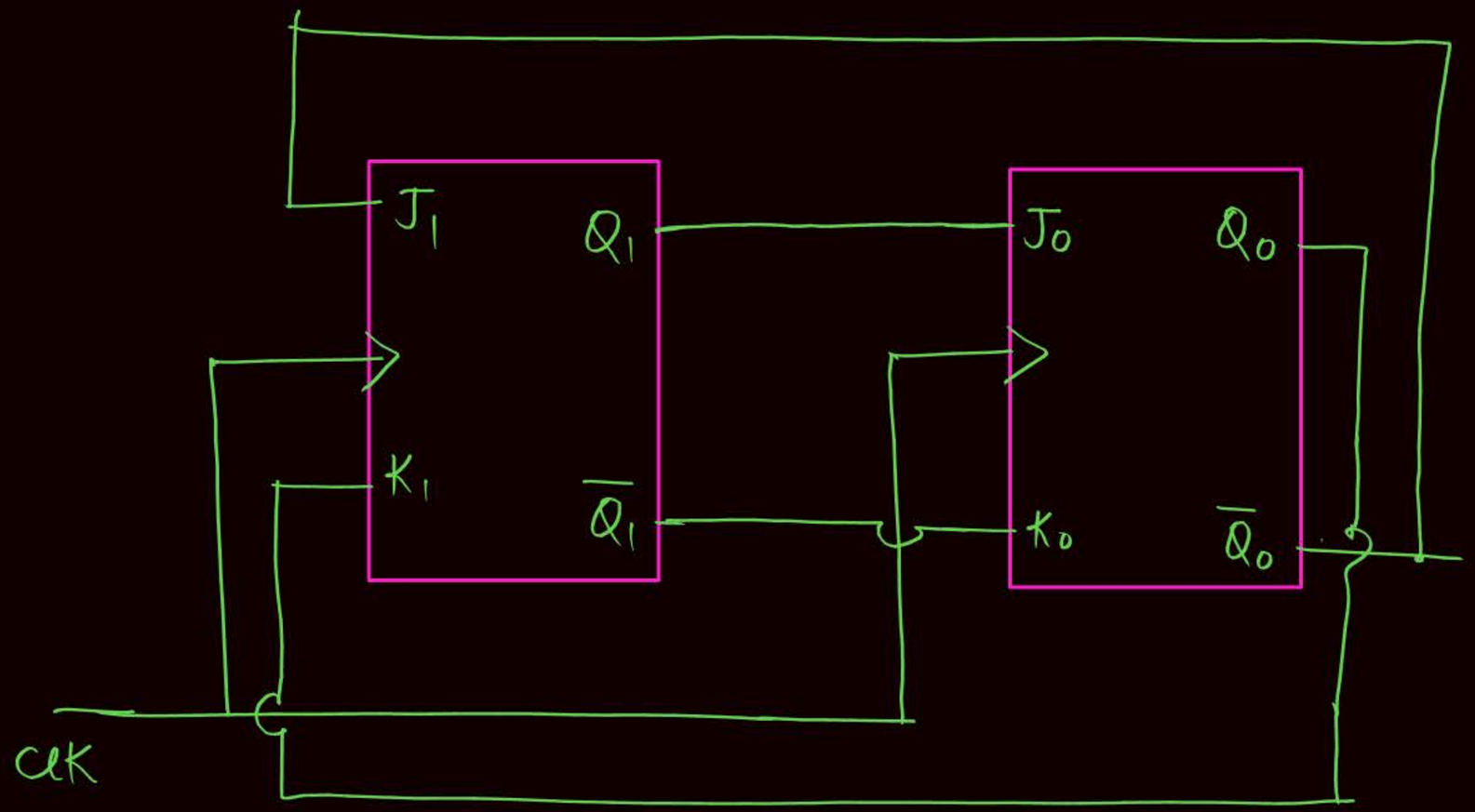


$\overline{J_0} = Q_1(n)$



$K_0 = \overline{Q_1}(n)$

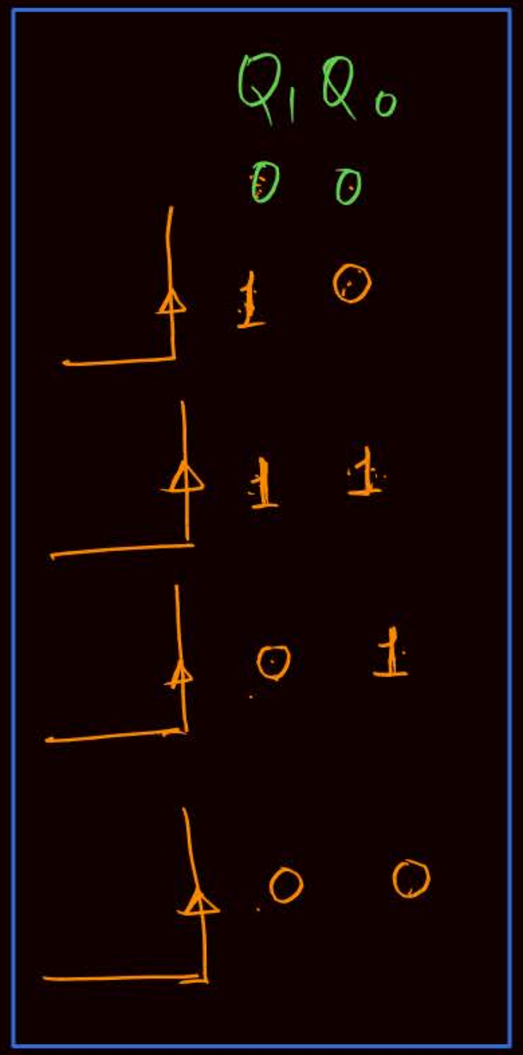
$Q_1 Q_0 = 0$

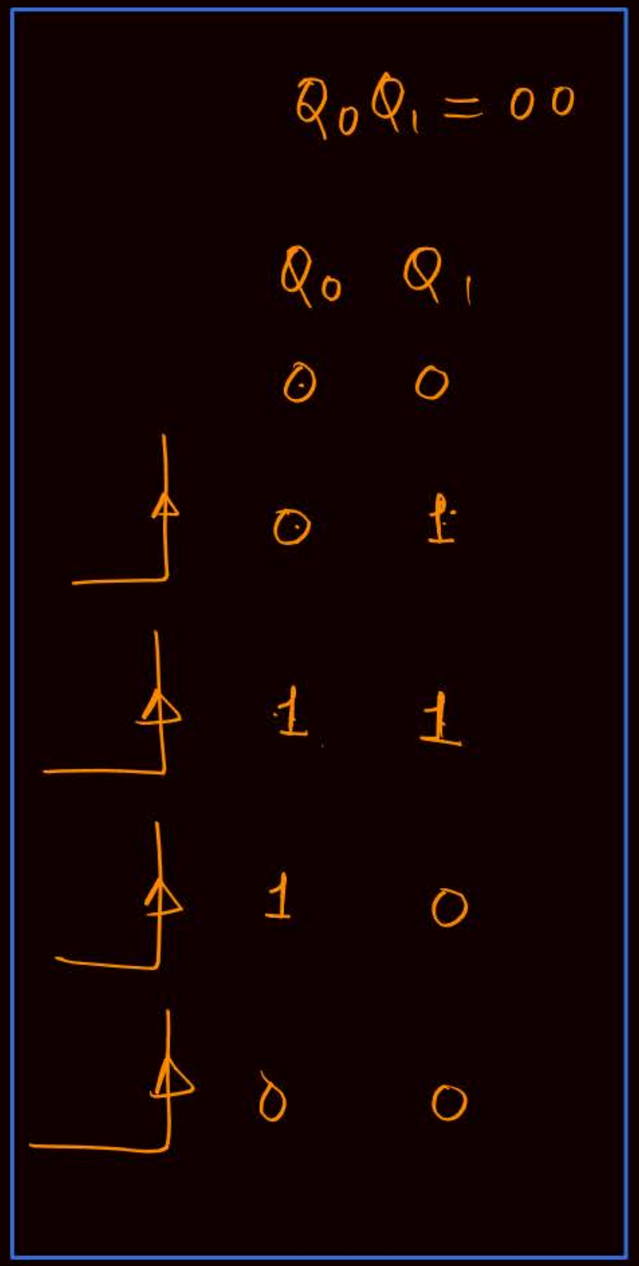


$$Q_1(n+1) = J_1\overline{Q}_1(n) + \overline{K}_1 Q_1(n) = \overline{Q}_0(n)\,\overline{Q}_1(n) + \overline{Q}_0(n)\,Q_1(n) = \overline{Q}_0(n)$$

$$Q_1(n+1) = \overline{Q_0(n)}$$

$$Q_0(n+1) = J_0\overline{Q}_0(n) + \overline{K_0}Q_0(n) = Q_1(n)\,\overline{Q}_0(n) + Q_1(n)\,Q_0(n) = Q_1(n)$$


$Q_1 Q_0$
0 0
1 0
1 1
0 1
0 0


$Q_0 Q_1 = 00$
Q_0 Q_1
0 0
0 1
1 1
1 0
0 0

H.W.

- Design a Counter with sequence:

1. $(0-1-4-2-5)$

Design using T-FF.

2. $(1-2-4-8-12-14-15-0-3)$ using D-FFs.

Q. Design a Counter that goes through states :

0 – 3 – 1 – 2 – 2

got saturated at '2' ⟶ saturated Counter.

using T-ff.



Topic : 2 Min Summary

→ Synch. Counter

Thank you

GW

Soldiers !

